Akshay (Ash) Deshmukh

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Analyst with 2+ years of experience delivering data-driven insights across public sector, retail, and commercial environments. Strong expertise in Python, SQL, Power BI, and Excel for data analysis, modelling, and visualisation. Proven track record of building scalable dashboards, automating reporting workflows, and applying statistical and machine learning techniques to solve business problems. **MSc Data Science and Statistics** graduate with a solid foundation in translating complex data into actionable insights for stakeholders.

CERTIFICATIONS and SKILLS

- Python
- Power BI
- SQL
- Excel
- Data Modelling
- Lean Six Sigma
- Dashboard Development
- AWS
- Data Visualisation
- Machine Learning
- Requirements Gathering
- Process Improvement
- Cross-Functional Collaboration
- Problem Solving
- Stakeholder Communication
- Critical Thinking
- Organisational Skills

WORK EXPERIENCE

Business and Data Analyst – Stoke-on-Trent Council *January 2025 – Present*

- Gather and translate business requirements from housing, finance, and operations teams into data-driven solutions, enabling more effective reporting and decision-making.
- Analyse large financial and operational datasets using SQL and Excel to identify trends, anomalies, and cost drivers, supporting budgeting and forecasting processes.
- Develop and maintain interactive Power BI dashboards tracking 15+ KPIs, reducing ad hoc reporting requests by 60% and enabling self-service analytics.
- Improve data quality and reporting accuracy by up to 20% through data validation, cleansing, and structured gap analysis.
- Optimise SQL queries and data models, improving dashboard performance and reducing load times by ~30%.
- Automate manual reporting workflows, saving approximately £10,000 annually and significantly reducing turnaround time.
- Manage multiple analytical projects simultaneously, prioritising based on business impact and ensuring timely delivery.

Data Scientist – Shreyas Technosoft *June 2023 - September 2023*

- Built demand forecasting models using time-series techniques (ARIMA, Prophet) on large-scale datasets, reducing inventory overhead by 18%.
- Developed automated data validation pipelines in Python, achieving 99.5% data accuracy across multiple data sources.
- Designed real-time data pipelines integrating cloud platforms, improving response time to market changes by 80%.
- Contributed to AI-driven solutions for content generation and optimisation in marketing workflows.

Data Science Intern – Shreyas Technosoft *September 2021 - May 2023*

- Designed and executed A/B and Bayesian experiments to optimise pricing strategies, increasing conversion rates by 13%.
- Built machine learning models for forecasting, segmentation, and classification, supporting revenue growth and strategic planning.
- Developed 5+ Power BI dashboards consolidating multi-source data, eliminating 35+ hours of manual reporting monthly.
- Automated data processing workflows using Python, improving efficiency and consistency across projects.
- Migrated Tableau dashboards to Power BI, reducing costs and improving reporting scalability.
- Delivered insights and presentations to senior stakeholders, translating complex analysis into clear business recommendations.
- Implemented MLflow and CI/CD pipelines, reducing model deployment time by 30% and improving reproducibility.

EDUCATION

MSc Data Science and Analytics, University of Leeds, UK *September 2023 – September 2024*

- **Merit | Russell Group |** Modules: Programming, Time Series, Regression, DL, ML

MSc Statistics, Pune University, India *August 2021 – May 2023*

- **Distinction |** Modules: Probability Theory, Bayesian Inference, Calculus, Linear Algebra, R, SAS.

INDEPENDENT PROJECTS

- **Dealing with multiple missing values (Masters) –** Analysed mobile features dataset, implementing various imputation methods to maintain data integrity.
- **Income Prediction (Guided) –** Implemented machine learning models to predict income, leveraging advanced analytical techniques and GitHub for streamlined project management.
- **Work-Life Balance (Post-Graduation) –** Conducted peer-reviewed primary data acquisition and statistical analysis on government and private sector employees, deriving actionable insights.
- **Loan Default Prediction (Independent)** – Built a classification model to predict borrower default risk by handling multicollinearity, imputing missing values, and engineering key features. Compared and fine-tuned multiple ML algorithms to identify the best-performing model for accurate risk assessment.
- **Banking Customer Churn Analysis (Independent)** – Analysed churn behaviour across 10k+ customer records to identify key attrition drivers. Built and compared ML models to predict churn probability, delivering a reliable classifier to support targeted retention efforts.